AuditGPT

Financial Analysis

TEAM : WE OVERSLEPT

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Problem Statement

- Corporate fraud signals exist in financial reports but remain unnoticed due to year-by-year analysis.
- Key red flags like cash flow mismatches, rising debt, and changing auditor language are missed over time.

Main Challenges

The challenge is to build a financial statement forensics engine that:

- Ingests a company's complete filing history
- Analyzes multi-year financial and textual data
- Detects hidden fraud patterns and anomalies
- Generates a fraud risk and financial health report

Our Solution

- **Data Collection:**
  10 years financial data using **yfinance + Screener**

- **Data Processing:**
  Cleaned & structured data → extract **revenue, cash flow, debt, ratios, RPT**

- **Anomaly Detection:**
  ML + Rule-based models to detect unusual patterns (cash flow mismatch, rising debt, spikes)

- **Industry Benchmarking (Core Idea):**
  Compare company with **peer companies in same sector** to identify what's normal

- **Context-Aware Analysis:**
  Flag only those anomalies that **deviate from industry norms**

- **Red Flag Generation:**
  Each flag includes:
  - Year, Statement type, Line reference
  - Severity (**Low / Medium / High / Critical**)

- **NLP Sentiment Analysis:**
  Analyze **auditor notes trend** (positive → negative)---

[User Input]
(Company Name / CIN)
↓
[Data Collection & Processing]
(yfinance + NSE + Screener APIs)
↓
[Feature Extraction]
(Revenue, Debt, Cash Flow, RPT, Ratios)
↓
[Industry Identification]
(Determine company sector)
↓
[Peer Selection]
(Select similar companies)
↓
[Industry Benchmarking]
(Compute peer averages & normal ranges)
↓
[Anomaly Detection]
(Compare:
- Historical trends & Industry benchmarks)
↓
[Red Flag Generation]
(Assign severity + add references)
↓
[NLP Sentiment Analysis]
(Auditor notes trend)
↓
[Insight Generation]
(Fraud Risk Score + Summary)
↓
[Dashboard Output]

Techstack

Backend / Framework:-

Python — core language
Flask – web framework

AI / LLM:-

Google Generative AI (Gemini) — google-generativeai Gemini API

Data & Finance:-

Pandas / NumPy — data manipulation
yfinance — Yahoo Finance market data
bselib — BSE (Bombay Stock Exchange) data

PDF Processing:-

pdfplumber / pdfminer.six / PyPDF2 — PDF parsing and text extraction

Web Scraping:-

BeautifulSoup4 / html5lib / lxml — HTML parsing Requests / curl_cffi — HTTP requests

Config & Utilities:-

python-dotenv — environment variable management Pydantic — data validation

Key Features

Auditor Sentiment Trend

Track positive, neutral, and negative sentiment across 10 years with visual trend lines

Fraud Risk Scoring

Automated anomaly detection flags sudden changes and red flags in auditor behavior

Related Party Analysis

Identify and track related party transactions that may indicate conflict of interest

Peer Benchmarking

Compare auditor sentiment and risk scores against industry peers

AI Explanations

LLM-powered plain language explanations make complex findings accessible to everyone

Custom Alerts

Set thresholds and receive notifications when risk scores exceed your parameters

Data Sources

BSE & NSE Filings

Annual reports from Bombay Stock Exchange and National Stock Exchange

Financial Statements

Balance sheets, profit & loss statements, and cash flow statements

Auditor Notes

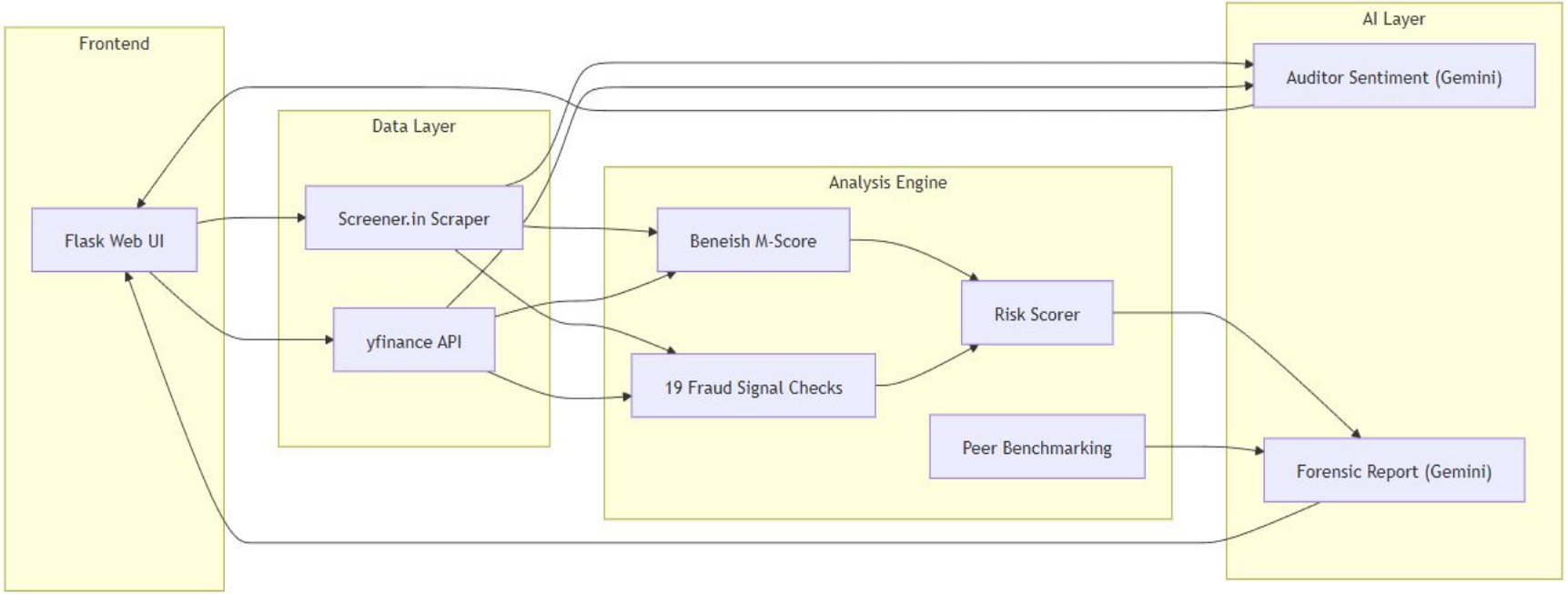
Detailed auditor observations and qualifications sections

Company Filings

Regulatory disclosures and supplementary documentation

Currently uses manual PDF scraping with plans for live API integration with exchange platforms

System Architecture



Core Logic

Core Logic – Fraud Risk Score

Step 1: Data Extraction
Fetch **10 years financial data** (yfinance + NSE)
Step 2: Feature Calculation
Compute key metrics:
Revenue, Cash Flow, Debt, Ratios
Step 3: Fraud Indicators
Apply:
Beneish M-Score
Financial anomaly rules
Step 4: Pattern Analysis
Detect patterns like:
Revenue > Cash Flow
Rising debt trend
Step 5: Risk Scoring
Assign score:
Low / Medium / High / Critical
Step 6: AI Explanation
Generate simple **fraud reasoning**

Core Logic – Red Flag Detection

Step 1: Identify Anomalies
Detect unusual values in metrics
Step 2: Context Check
Validate using **industry benchmarks**
Step 3: Flag Generation
Convert anomalies → red flags
Step 4: Attach Details
Include:
Year
Statement type
Line item reference
Step 5: Severity Assignment
Low / Medium / High / Critical

Core Logic

Core Logic – Red Flag Timeline

Step 1: Collect Flags
Gather all detected anomalies
Step 2: Year Mapping
Assign each flag to a specific year
Step 3: Trend Tracking
Track how issue evolves over time
Step 4: Visualization
Create **timeline view (year-wise progression)**

Core Logic – Anomaly Map

Step 1: Metric Selection
Choose key metrics (Revenue, Debt, Cash Flow, RPT)
Step 2: Year-wise Analysis
Analyze values across 10 years
Step 3: Deviation Detection
Compare:
Historical trend
Industry benchmark
Step 4: Severity Mapping
Assign:
Normal / Warning / Critical
Step 5: Heatmap Generation
Visualize as **color-coded grid**

Core Logic

Core Logic – Industry Peer Comparison

Step 1: Identify Industry
Detect company sector
Step 2: Select Peers
Choose 5+ similar companies
Step 3: Data Collection
Fetch peer financial metrics
Step 4: Benchmark Calculation
Compute averages & ranges
Step 5: Comparison
Compare company vs peers
Step 6: Insight Generation
Highlight deviations with context

Core Logic – Auditor Sentiment Analysis

Step 1: Extract
Identify auditor sections from reports
Step 2: Clean
Remove noise and irrelevant text
Step 3: Sentiment Analysis
Classify tone:
Positive / Neutral / Negative
Step 4: Trend Analysis
Track sentiment over 10 years
Step 5: Risk Detection
Detect sudden negative shifts
Step 6: AI Explanation
Convert into simple insights

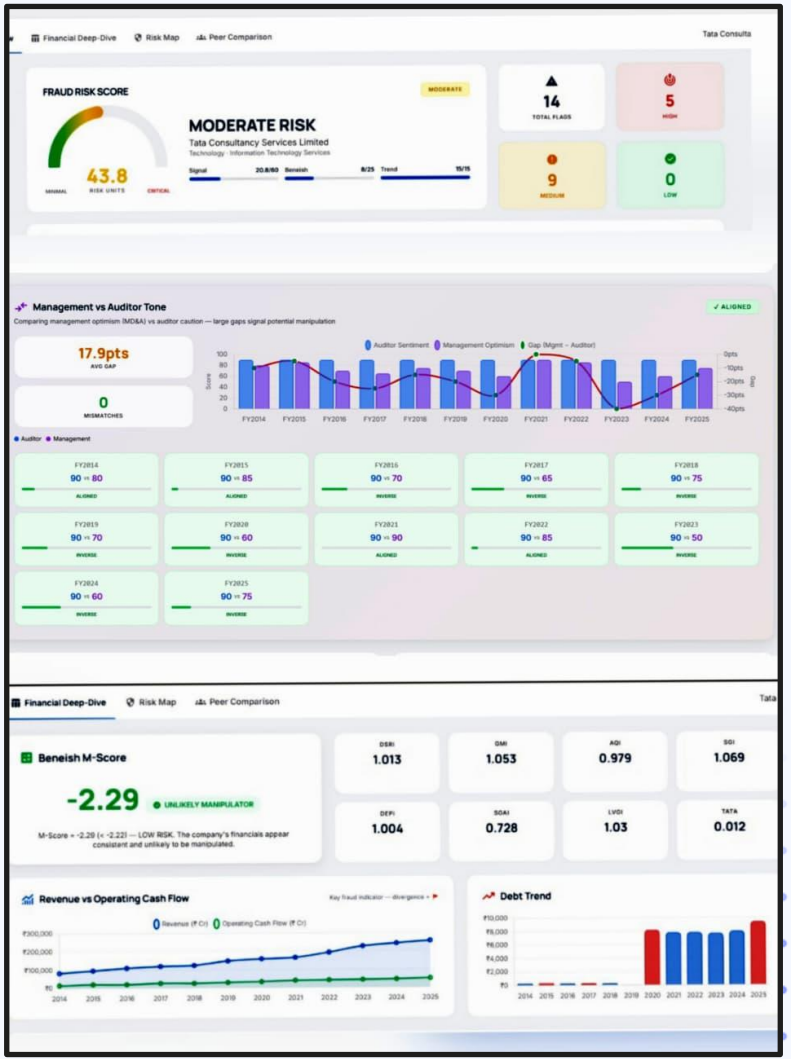
Prototype

Key Metrices:-

- Fraud risk score (0 - 100)
- Sentimental Analysis
- Anomaly Detection
- Related Party Transaction tracking
- Management vs Auditor tone

Visual Trends:-

- Sentimental Trend Graphs
- Risk Score
- Revenue vs CashFlow
- Debt Trend
- Anomaly Deviations



IT Peers

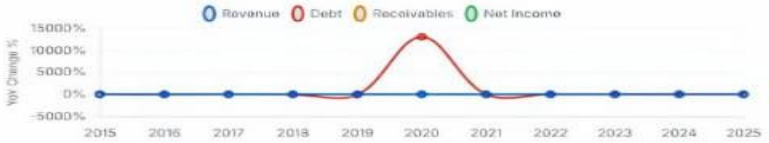
Peers: INFY.NS, WIPRO.NS, HCLTECH.NS, TECHM.NS, LTI.NS

Benchmarked against 5 live IT peers.

COMPANY	PM	D/E	ROA	ROE
TCS	19.11	0.08	27.85	43.83
INFY	16.15	0.11	15.67	32.68
WIPRO	14.59	0.22	6.77	15.2
HCLTECH	13.02	0.1	12.87	23.19
TECHM	8.39	0.07	—	—
LTI	—	—	—	—

Anomaly Deviation Map

YoY % change in key metrics — large spikes signal potential manipulation



Red Flag Timeline

Forensic anomalies detected across financial years

Critical · Medium · Low · 14 flags

HIGH — FY2025
Debt Growing Faster Than Revenue
Debt grew 17.1% vs revenue growth of 6.0%. Company may be funding operations/growth through unsustainable borrowing.

MEDIUM — FY2024
Gross Margin Compression
Gross margin dropped from 55.1% to 45.5% over 2 years. Sustained margin compression indicates pricing pressure or cost issues.

MEDIUM — FY2024
SGA Expense Anomaly
SGA/Revenue dropped from 8.0% to 5.8%. Sharp SGA decline may indicate improper expense capitalization.

MEDIUM — FY2023
Inventory Buildup
Inventory grew 40.0% vs COGS growth of 10.9%. Excess inventory may indicate demand weakness or obsolescence risk.

MEDIUM — FY2023
Capex Cuts During Growth
Revenue grew 17.6% but capex declined -25.7%. Cutting investment during growth may inflate short-term profits unsustainably.

MEDIUM — FY2023
Gross Margin Compression
Gross margin dropped from 55.7% to 47.9% over 2 years. Sustained margin compression indicates pricing pressure or cost issues.

MEDIUM — FY2023
SGA Expense Anomaly
SGA/Revenue dropped from 12.3% to 8.0%. Sharp SGA decline may indicate improper expense capitalization.

HIGH — FY2022
Revenue-Cash Flow Divergence
Revenue grew 16.8% but operating cash flow grew only 3.0%. Healthy companies should see cash flow track revenue growth.

MEDIUM — FY2022

HIGH — FY2020

Future Scope and Impact

AuditGPT can help detect fraud early reduce the time required for financial analysis, and assist investors and auditors in making better decisions.

The system can be extended to support real-time monitoring, integration with regulatory platforms, and analysis of global companies.

AuditGPT analyzes 10 years of finanacial data the way a forensic accountant would, but does it in under 90 seconds using AI.

Git-Hub Link:-

https://github.com/MadManav/auditgpt